

Data Analytics Assessment – Completed Guide (Q1–Q4)

Dataset: UCI Online Retail Dataset

Important: the screenshots supplied in the chat show Q1–Q4 only. Pages containing Q5–Q6 were not supplied, so this pack does not invent those requirements.

Q1 – Dataset Selection

Field	Answer
Dataset Name	Online Retail Dataset
Source	UCI Machine Learning Repository
Official URL	https://archive.ics.uci.edu/dataset/352/online+retail
Records	541,909
Columns	8
Period	1 December 2010 – 9 December 2011

Brief description: The dataset contains transactions from a UK-based registered non-store online retailer. The company mainly sells unique all-occasion gift products and many customers are wholesalers. It includes invoices, products, quantities, dates, unit prices, customer IDs and countries.

Why selected: It is large, credible, publicly available and complex enough for meaningful sales, customer, product and geographical analysis.

Business opportunities: Product performance, revenue trends, customer value, country performance, cancellations/returns and inventory/marketing decisions.

Q2 – Business Understanding

Item	Answer
Business Problem	Understand key drivers of sales and revenue and identify opportunities to improve sales, customer retention, product strategy and geographical performance.
Question 1	Which products generate the highest sales revenue?
Question 2	Which countries generate the highest revenue and sales volume?
Question 3	How does revenue change over time, particularly across different months?
Question 4	Who are the highest-value customers based on total spending and purchase frequency?
Question 5	What proportion of transactions are cancelled, and which products or customers are associated with cancellations?

Hypothesis	Statement
H1	A relatively small group of high-value customers contributes a disproportionately large share of total revenue.
H2	A small number of products generate a disproportionately large share of total revenue.

Hypothesis	Statement
H3	A small number of countries contribute a large proportion of total revenue.
H4	Revenue varies across months, with some months generating substantially higher sales.

Q3 – Data Processing

Change	Why necessary?	If not done
Missing CustomerID	Retain useful transaction records while marking unknown customers.	Large amount of sales data could be lost and customer analysis biased.
Missing Description	Keep records and mark missing descriptions clearly.	Valid transactions could be removed.
Remove duplicates	Prevent repeated transaction lines from being counted twice.	Revenue/counts could be overstated.
Datetime conversion	Enable reliable time-based analysis.	Monthly/yearly analysis becomes difficult.
Revenue field	Quantity × UnitPrice creates a consistent monetary measure.	Repeated manual calculations would be required.
Date fields	Year, Month, Month_Name, Year_Month support trend analysis.	Time trends would be harder to analyse.
Cancellation flag	Invoice numbers beginning with C indicate cancellations.	Sales could be overstated if cancellations are mixed with sales.
Outlier review	Extreme values may be legitimate wholesale orders or returns.	Valid business activity could be incorrectly deleted.

Q4 – Five Business Insights

Insight	Evidence to report	Business impact	Recommendation
Top products	Top 10 by aggregated Revenue/Quantity.	Strong products drive revenue and inventory needs.	Protect stock and promote strong products.
Country concentration	Top countries by Revenue and Transactions.	Shows key markets and concentration risk.	Prioritize strong markets and test promising smaller markets.
Monthly trend	Highest/lowest Year_Month by Revenue.	Seasonality affects planning.	Prepare inventory and marketing before peaks.
High-value customers	Top customers by Revenue and frequency.	Retention of valuable customers protects revenue.	Use targeted loyalty/retention actions.
Cancellations/returns	Cancellation rate and cancellation totals.	May indicate product/fulfilment/service issues.	Investigate cancellation drivers by product/customer.

Execution note

The accompanying Excel workbook contains the Q1–Q4 answers and the supplied Google Colab notebook contains the Python processing steps. Upload the official Online Retail.xlsx dataset to Colab, run the notebook, and paste the resulting processed data into the Processed Data worksheet. Q4 should use the actual outputs from those calculations rather than invented figures.